Physical Cume
September 14, 2001

In molecular electronic spectroscopy:

1.) Regarding the Franck-Condon principle:

a) Define the Franck-Condon principle and describe the processes induced in a molecule by a molecular electronic transition.
b) Derive the Franck-Condon principle from the *transition dipole moment*.
c) What is the *Franck-Condon factor* and what does it identify as an important factor in determining absorption intensity in a molecular electronic transition?

2.) Describe the sequence of steps leading to fluorescence.

3.) Describe the sequence of steps leading to phosphorescence.

4.) Describe the steps involved in a) dissociation and b) predissociation.

5.) a) In what way does the information contained in the vibrational structure observed in the absorption spectrum of a given molecule differ from the information contained in the vibrational structure observed in the fluorescence spectrum of that same molecule? (Draw a simple figure showing a typical absorption spectrum and the corresponding fluorescence spectrum in a graph in which wavelength is plotted on the x-axis, and intensity on the y-axis.)

b) Is it possible to find transitions which are coincident in the absorption spectrum and fluorescence spectrum of a molecule? Explain.

Credits: no.1, 40 points; nos. 2, 3, 4a, 4b, 5a and 5b, 10 points each.